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A Literature Survey on Cell Tracking in Time-Lapse Microscopy

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Abstract

Cell tracking is a key task in computational biology and bioimaging, aimed at studying cellular dynamics such as motility, mitosis, and interaction over time. Time-lapse microscopy provides sequences of images that capture these processes, but automatic tracking of cells is challenged by overlapping, morphological variability, and movement complexity. This literature survey explores traditional and machine learning-based approaches to cell tracking, covering segmentation-tracking pipelines, deep learning models, and recent benchmarks.

Keywords: Cell tracking · Time-lapse microscopy · Deep learning · Object tracking · Biomedical image analysis · Instance segmentation

1 Introduction

Tracking cells in microscopy videos is crucial for understanding biological phenomena such as embryonic development, cancer metastasis, and immune responses. Cell tracking involves not only segmenting cells at each time point but also maintaining their identities across frames. While manual tracking is time-consuming and prone to error, automated methods face challenges due to cell division, appearance changes, and occlusions.

This survey reviews various methods used for cell tracking, from rule-based algorithms to modern deep learning frameworks, emphasizing accuracy, scalability, and generalization.

2 Related Work

Traditional Methods

Traditional approaches often treat tracking as a data association problem based on features like centroid, area, and velocity.

Nearest Neighbor and Kalman Filtering: Basic methods associate detected cells frame-to-frame using distance metrics or Kalman filters to predict positions [1].

Graph-Based Tracking: These methods construct graphs over time where nodes represent cell instances and edges encode probable transitions [2].

Challenges: Classical models struggle in dense regions and with cell division events (mitosis), requiring hand-crafted heuristics.

Deep Learning Approaches

DeepCell and U-Net Extensions: Some approaches first segment each frame using CNNs like U-Net, then apply linking algorithms such as the Viterbi algorithm to associate segments [3].

Recurrent Neural Networks (RNNs): RNNs and LSTM networks have been proposed to model temporal dependencies in cell behavior [4].

End-to-End Tracking Models: Newer models, such as Tracking-by-Detection frameworks and transformer-based architectures, jointly learn spatial and temporal features for improved accuracy in crowded scenes.

Benchmarks and Competitions

Cell Tracking Challenge (CTC): The CTC provides diverse annotated datasets and evaluation metrics for benchmarking. Metrics include Tracking Accuracy (TRA), **Detection Accuracy (DET)**, and **Segmentation Accuracy (SEG)**.

3 Challenges and Future Work

While progress is evident, cell tracking remains difficult due to:

- Mitosis detection and lineage tracking
- Tracking in 3D and across multiple focal planes
- Variation in staining and imaging modalities
- Limited generalization across cell types

Future directions include:

- Self-supervised learning and domain adaptation
- Real-time cell tracking in live imaging setups
- Integrating biological priors into learning models

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